

MANASA REDDY SEELAM

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WORK EXPERIENCE

SUMMARY

- Over 3 years of experience in Data Engineering and Proficiency in **Python**, **SQL**, and **R** programming languages for data manipulation, analysis, and visualization, with extensive experience in utilizing libraries such as **NumPy**, **Pandas**, and **Matplotlib**.
- Experienced in managing and optimizing databases including **MySQL**, **Oracle**, **MongoDB**, and **SQL Server**, with expertise in designing and implementing efficient data models and schemas.
- Skilled in cloud technologies, particularly AWS services such as **EC2**, **S3**, **RDS**, **Lambda**, **CloudWatch** for building scalable and reliable data processing pipelines.
- Proficient in visualization tools like **Tableau**, **Power BI**, and **MS Excel** for creating insightful dashboards and reports to communicate key findings and insights to stakeholders.
- Experienced in big data tools including **Hadoop**, **Apache Spark**, **Hive**, **Pig**, **MapReduce**, and **Kafka** for processing and analyzing large volumes of structured and unstructured data.
- Familiar with **ETL** (Extract, Transform, Load) tools such as **Informatica PowerCenter**, **Talend**, **SSIS**, **SSRS**, and **SSAS** for data integration, cleansing, and transformation tasks.
- Well-versed in **Agile** (Scrum) and Waterfall methodologies for project management and collaboration, with experience in working within cross-functional teams to deliver high-quality data solutions.

BCBS

Data Engineer

TN, USA

Feb 2024 - Present

- Utilized **SQL** and **Python** to develop complex **ETL** processes, improving data integrity and consistency across multiple healthcare data sources, resulting in a significant reduction in data discrepancies and faster processing of patient records and clinical data.
- Leveraged **SQL Server** and **MySQL** databases to store and manage structured healthcare data, optimizing database performance, indexing strategies, and query optimization techniques to ensure swift retrieval of patient information and medical records.
- Designed and implemented scalable data pipelines using **Hive** and **MapReduce**, enabling efficient processing of large-scale healthcare datasets, significantly increasing data throughput and reducing processing time for medical research and clinical studies.
- Implemented machine learning models using libraries like **Scikit-Learn**, integrating predictive analytics capabilities into healthcare data processing pipelines to derive valuable insights from historical patient data, aiding in disease prediction and personalized treatment plans.
- Developed and maintained data warehousing solutions using **Microsoft SQL Server Analysis Services (SSAS)**, supporting the creation of efficient data cubes and multidimensional analysis, boosting report generation efficiency for healthcare outcomes and operational performance to an average of 2 hours per report.
- Designed and developed interactive, user-friendly dashboards and visualizations in **Tableau**, incorporating advanced charting techniques, calculations, and data blending to present complex healthcare data in an intuitive manner for clinicians and administrators.
- Implemented **Excel VBA** macros to automate repetitive healthcare data processing tasks, streamlining workflows and significantly reducing manual processing effort for tasks such as patient billing, appointment scheduling, and compliance reporting.
- Leveraged **Jupyter Notebook** as an integrated development environment (**IDE**) for healthcare data exploration, prototyping, and sharing reproducible analysis with medical researchers and healthcare stakeholders.

Accenture Solutions Pvt Ltd

Application Development Associate / Data Engineer

Chennai, India

Sep 2020 – Apr 2022

- Utilized **Python** for scripting and automation of data processes, incorporating **Matplotlib** to generate custom data visualizations, which accelerated data analysis processes by completing tasks in half the time.
- Developed and executed complex **SQL** queries across **MongoDB** and **SQL Server** to facilitate efficient data extraction and management, supporting large-scale data applications.
- Managed and optimized **ETL** processes using **Informatica PowerCenter** and **Talend**, streamlining data integration and transformation workflows, which reduced processing time from disparate sources to centralized repositories by 2 hours per batch.
- Designed and implemented data storage and warehousing solutions on **AWS** using services like **Glue** and **RedShift**, enhancing data availability and query performance.
- Monitored and audited cloud environments with **AWS CloudWatch**, ensuring compliance with data security standards and operational best practices, resulting in improved security incident management.
- Leveraged big data technologies such as **Hadoop** and **Apache Spark** to process and analyze voluminous datasets, improving data processing speed and scalability.

- Deployed and managed **Power BI** solutions on the Power BI Service, ensuring secure data access and role-based permissions, enhancing report access speed and integration efficiency.
- Automated report generation processes by leveraging Excel's **VBA** (Visual Basic for Applications) to create macros and user-defined functions, streamlining data analysis and reporting tasks.
- Adhered to the Waterfall methodology to manage data engineering projects, ensuring structured and sequential project delivery that met predefined project milestones.
- Utilized **GitHub** for version control, enhancing team collaboration and maintaining code quality, resulting in reduced bug rates throughout project cycles.

Nokia

Data Analyst

India

Sep 2019 – Aug 2020

- Utilized Python to create and maintain scripts for data processing and analysis, leveraging libraries such as NumPy and Pandas to handle large datasets efficiently.
- Developed complex **SQL** queries to extract and manipulate data from **MySQL** databases, enhancing the accuracy and relevancy of data for analysis, which resulted in reducing data retrieval time from 30 minutes to 10 minutes.
- Managed and optimized Extract, Transform, Load (**ETL**) operations to ensure accurate and efficient data processing, reviewing and refining data integration strategies to enhance throughput and data quality, significantly reducing processing times and errors
- Designed and developed intuitive and user-friendly **Power BI** dashboards that provide critical insights and analytics for decision-makers, focusing on interactive and visually appealing elements to facilitate informed decision-making across the organization
- Implemented and managed automation solutions using Power Automate to streamline business processes and reduce manual effort, developing automation scripts that improved operational efficiency by over 30%
- Regularly engaged with department managers and other key stakeholders to gather and analyze data requirements, provided consistent updates on project progress and data insights to ensure alignment with business objectives and data-driven decisions
- Applied Agile (**Scrum**) methodologies to manage and execute data projects, ensuring timely delivery and adherence to business requirements through regular sprint reviews and planning sessions.
- Conducted data cleansing and transformation to improve data quality and consistency, which supported accurate and effective data analysis, reducing data-related errors by half.

PROJECT EXPERIENCE

Driver drowsiness detection System

- The project is designed to overcome the problem of accidents due to drowsiness.
- This system will alert the driver while driving based on the number of times eyebrows close per minute with the help of a camera.

Report creation of data using Power BI and Automating tasks using Power Automation

- Generated reports with complex data along with DAX formula usage. On the other hand, I have automated workflows for many scenarios.

Density based Traffic Controlling System

- The aim of the project is to monitor the patient's health condition from a mobile place without staying beside the patient.
- It will record the readings from time to time and intimates if there is any abnormal condition.

EDUCATION

Christian Brothers University - Memphis, TN

Masters of Science in Computer Information Systems

Jun 2022 – Dec 2023

Veltech University

Bachelors of Technology in Computer Information Systems

Jun 2016 – May 2020

SKILLS & INTERESTS

Skills: Python, SQL, PowerShell, VB Script, MS SQL Server, Oracle, MongoDB, Snowflake, PostgreSQL, Teradata, PySpark, SparkSQL, Pandas,, NumPy, Matplotlib, Seaborn, Hadoop, MapReduce, Hive, Apache Spark, Pig, Airflow, Kafka, HDFS, Yarn, Informatica (IICS), DataStage,, SSIS, Agile, Waterfall, Tableau, Power BI, Excel, Git, JIRA, Jenkins,, Visual Studio, Jupyter Notebook, PyCharm, Azure (Data Factory, Databricks, Data Lake Storage, Azure Stream Analytics, Azure Synapse, Azure Blob Storage), AWS (RDS, Lambda, S3, EventBridge, Glue, Athena). SSIS, SSMS, SSRS, SAS, OpenCV,SQL Server Apache, Airflow, SharePoint, Informatics, CI/CD, Linux/Unix, Apache Kafka, Apache Pig, SciPy

Soft Skills: Critical Thinking, Communication Skills, Presentation Skills, Problem-Solving

Data & Domain: Software Development, Start-ups, Data Analysis, Data Science, Data Engineering, Business Analysis, BI Analysis, Healthcare & Financial Dataset